

Units

ME 1411 – Material Science

Course Proficiencies

- Homework and In class 15%
- Lab Modules 25%
- Exams 30%
- Proficiencies 30%
 - Proficiency 1: Units 10%
 - Proficiency 2: The stress strain curve 10%
 - Proficiency 3: Stress and Strain Calcs 10%

From syllabus: Must master all three proficiency skill to pass this course.
Mastery is defined as earning at least a 90% on the question.

The first proficiency assessment is Tuesday

- 1st half of class: Module 3 in-class discussion
- 2nd half of class: Assessment

P2: Stress/Strain Curve

Question Type: Given a stress strain curve, determine the parameters

Examples Problems:

- HW 5 – Properties practice Question 1 and 2
- 5 – CE – Stress Strain Curve (p. 13 – 23)
- P2 – review test problems

Lectures:

- 05 – The Stress Strain Curve
- 06 – More Elastic properties (review slides)

P3: Stress Strain Calculations

Question Type: Open ended calculations.

Examples Problems:

- 5 – CE – Stress Strain_Curve (p. 14)
- 6 – CE – More Elastic Properties
- HW 6 – Units and Stress Calcs
- P3 test review problems

Lectures/Lab:

- 05 – The Stress Strain Curve
- 06 - More Elastic Properties
- 07 – Units

P1: Units

Question Type:

- Report units for variables in SI and US systems
- Convert units from SI to US, nm to mm, etc

Examples Problems:

- HW6 – Unit questions
- P1 Unit Test review questions

Lectures/Lab:

- 07 - Units

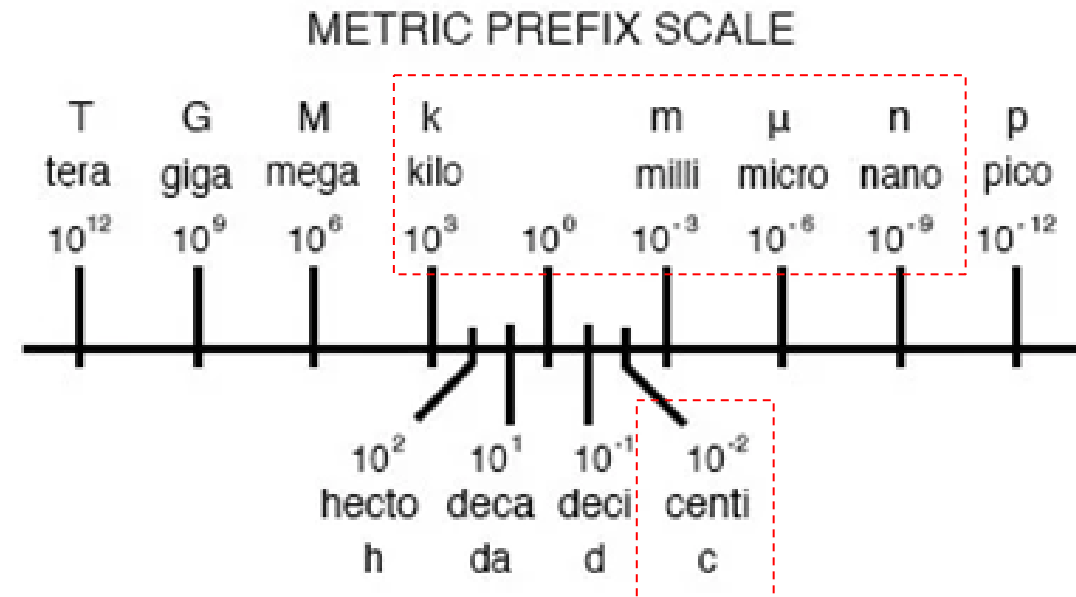
Unit Review

Length:

- Measures distance

- SI Unit: meter (m) 

- US unit:
 - Inch (in)
 - Feet (ft) = 12 in
 - Miles (mi) = 5280 ft



Converting Length Units

How many micrometers is 0.625 mm?

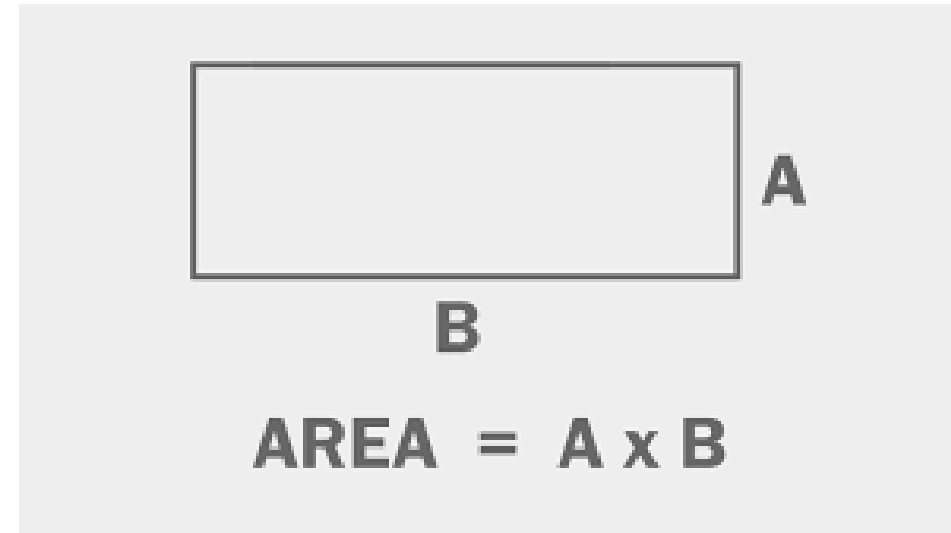
$$0.625 \text{ mm} \left(\frac{1000 \text{ } \mu\text{m}}{1 \text{ mm}} \right) = 625 \text{ } \mu\text{m}$$

How many inches is 3.65 cm?

$$3.65 \text{ cm} \left(\frac{1 \text{ in}}{2.54 \text{ cm}} \right) = 1.44 \text{ in}$$

Area

- SI units:
 - m^2
 - cm^2
 - mm^2
- US units:
 - in^2



Converting Units of Area

How many m² is 2840 mm²?

$$2840 \text{ mm}^2 \left(\frac{1 \text{ m}}{1000 \text{ mm}} \right)^2 = 0.00284 \text{ m}^2 = 2.84 \times 10^{-3} \text{ m}^2$$

How many square inches is $2.84 \times 10^{-3} \text{ m}^2$?

$$2.84 \times 10^{-3} \text{ m}^2 \left(\frac{1000 \text{ mm}}{1 \text{ m}} \right)^2 \left(\frac{1 \text{ in}}{25.4 \text{ mm}} \right)^2 = 4.40 \text{ in}^2$$

Volume

- SI units:

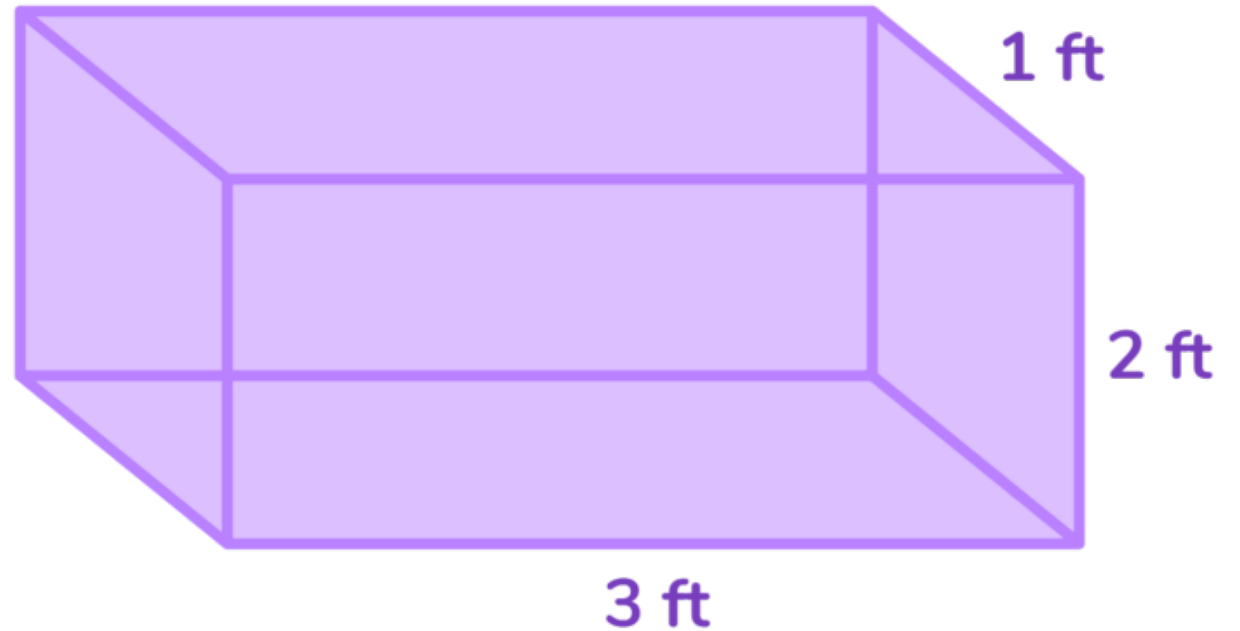
- m^3

- cm^3

- mm^3

- US units:

- in^3



Converting Units of Volume

How many m³ is 33040 mm³?

$$33040 \text{ mm}^3 \left(\frac{1 \text{ m}}{1000 \text{ mm}} \right)^3 = 3.30 \times 10^{-5} \text{ m}^3$$

How many cubic inches is $3.30 \times 10^{-5} \text{ m}^3$?

$$3.30 \times 10^{-5} \text{ m}^3 \left(\frac{1000 \text{ mm}}{1 \text{ m}} \right)^3 \left(\frac{1 \text{ in}}{25.4 \text{ mm}} \right)^3 = 2.02 \text{ in}^3$$

Force

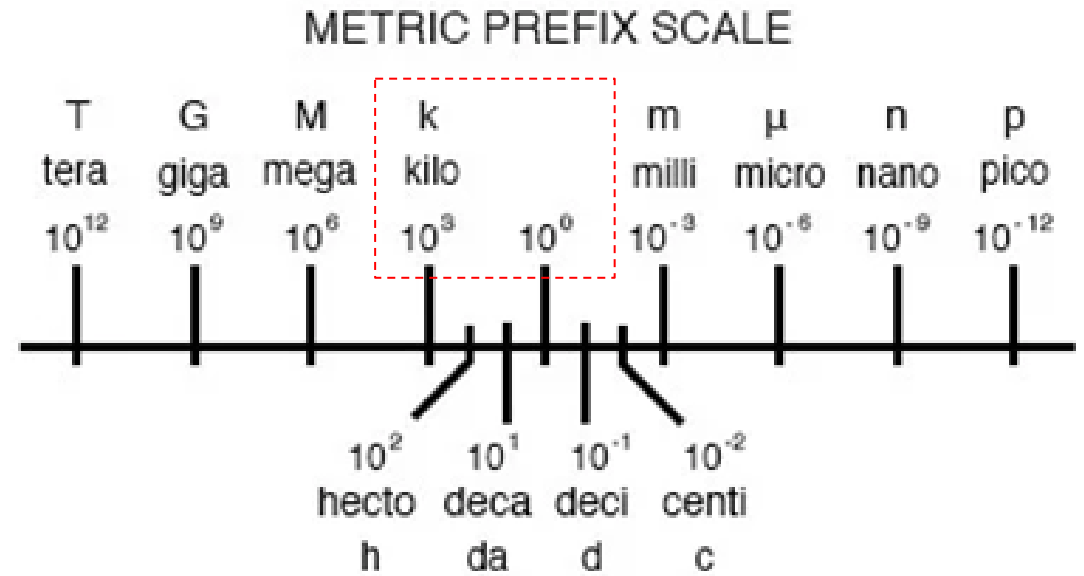
- SI Units: Newtons (N):

- $1 \text{ N} = \frac{\text{kg}}{\frac{\text{m}}{\text{s}^2}}$

- US Units: Pounds (lb)

- $1000 \text{ lb} = 1 \text{ kip}$

- For conversions: $1 \text{ lb} = 4.45 \text{ N}$



Converting Units of Force

How many pounds is 250 N?

$$250 \text{ N} \left(\frac{1 \text{ lb}}{4.45 \text{ N}} \right) = 56.2 \text{ lb}$$

How many Newtons is 820 lb?

$$820 \text{ lb} \left(\frac{4.45 \text{ N}}{1 \text{ lb}} \right) = 3650 \text{ N} = 3.65 \text{ kN}$$

Stress

- SI Units: Pascals (Pa)

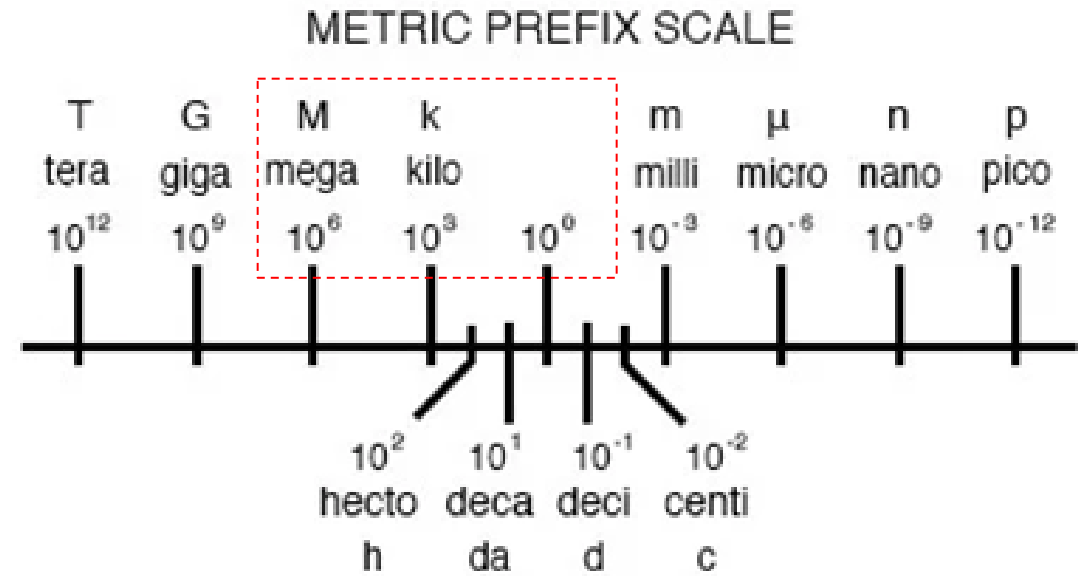
- Pa, kPa, Mpa

Commonly:

$$\text{Pa} = \text{N/m}^2$$

$$\text{kPa} = \text{kN/m}^2$$

$$\text{MPa} = \text{N/mm}^2$$



- US Units: pounds per square inch

- psi 1000 psi = kpsi

What units do we have?

For a tensile test, a 5 kN force is applied to a sample with cross-sectional area of 0.7854 cm² (1 cm diameter). What is the normal stress applied?

$$\sigma = \frac{P}{A}$$

****Use N for force. Use m for area. This will result in Pa ****

$$\sigma = \frac{5000 \text{ N}}{(0.7854 \text{ cm}^2) \left(\frac{1 \text{ m}}{100 \text{ cm}} \right)^2} = 6.37 \times 10^7 \text{ Pa} = 63.7 \text{ MPa}$$

****Note: Could use mm for area. N/mm² will result in MPa ****

$$\sigma = \frac{5000 \text{ N}}{(0.7854 \text{ cm}^2) \left(\frac{10 \text{ mm}}{1 \text{ cm}} \right)^2} = 63.7 \text{ MPa}$$

Strain

- Strain is unitless
- Sometimes we use $\mu\epsilon$ to represent 10^{-6}

Converting strain to microstrain:

$$0.000521 \rightarrow \text{Multiply by } 10^6 \rightarrow 521 \mu\epsilon$$

Converting microstrain to strain:

$$3405 \mu\epsilon \rightarrow \text{Multiply by } 10^{-6} \rightarrow 0.003405$$

Elastic Modulus

- SI Units: $\text{GPa} = 10^9 \text{ Pa}$
- US Units: $\text{Mpsi} = 10^6 \text{ psi}$

Hooke's Law + Units

$$\sigma = E \varepsilon$$

- Stress (sigma) and Modulus (E) have the same units because strain has no units
- Recommended: convert units to Pa or psi before calculating to avoid errors.

For the rest of class

- Work on HW6 – Units and calculations practice.